

Towards a Strategic Theory of the Firm:

Part II

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Over a decade ago, in the Fall of 1981, there was a small conference at the University of Southern California entitled "Non-Traditional Approaches to Policy Research." I prepared a paper for that conference entitled "Towards a Strategic Theory of the Firm" [Rumelt, 1984]. What was then fairly non-traditional about my approach was the use of microeconomic reasoning to try to capture strategic phenomena—entrepreneurship, innovation under ambiguity, industry evolution, and sustained heterogeneity despite competition. The contribution which seems to have had some staying power was my introduction of the term "isolating mechanisms" to denote phenomena, especially at the business level, which prevent the equilibration of returns. At that time, the dominant economic paradigm was the theory of entry barriers and mobility barriers, which create rents for collectives of firms by reducing entry. My notion was that there are additional impediments to equilibration rooted in the positions, assets, and histories of individual firms or even parts of firms. These isolating mechanisms include product reputations, learning and special know-how that is tacit and difficult to imitate, network externalities associated with de-facto standards, a host of first-mover advantages, etc. Since then, the concept of isolating mechanisms has become part of the evolving literature called the resource-based view of the firm.¹

The resource-based view has been a major theme in the economization of business and corporate strategy.² Today strategy researchers work with an amalgam of economic and quasi-economic reasoning. We envision the firm as striving to maximize value, but also see it as working with factors of production that are far from mobile, as dealing with ambiguous production functions, and as possessing or controlling collections of tacit knowledge and externally held attributions (reputation) that evolve over time in response to investment, activity and imitation.

It is useful to note that that this new view has *not* been developed by "applying" economics to strategy. Rather, it has been accomplished by carefully identifying the assumptions within received economic models that prevented or ruled out strategic phenomena, and by then analyzing the situations created by altered assumptions. In particular, a central dogma of neoclassical microtheory, especially that part associated with industrial organization, was the basic homogeneity of firms (but for scale). Differences in performance were thus attributed to differences in scale or its collective equivalent, concentration. By contrast, the resource based view has as its central dogma the heterogeneity of firms induced by heterogeneous resources.

The power behind this research stream has, in large measure, come from the clarity and strength of the theory (neoclassical microeconomics) which it attacks. In

¹Wernerfelt, Barney, Teece, Rumelt.

²RST Special Issue

fact, the neoclassical model maintained such a grip on the minds of economists that they often spoke of its failures as "market failures," as if it were somehow the responsibility of reality to live up to theory rather than the theorist's responsibility to describe reality.

One problem with this research strategy is that the foil, neoclassical microtheory, has died. Game theory now provides the basis for reasoning in industrial organization economics; agency theory provides explanations for non-optimal firm behavior, and transaction-cost economics describes the limits of markets, and the advantages of hierarchy, in carrying out coordination and exchange. One might argue that this is fine—the resource based view together with revived microtheory is the strategic theory of the firm. However, I believe that despite the flowering of a rich new microtheory that captures a great many strategic situations, we have dealt with only one-part of the story. There are a number of erroneous assumptions that most economically-oriented strategy researchers continue to borrow from economics. At this moment those that are clearest are plasticity, rationality of action, and homogeneity of beliefs. I believe that the most important of these is plasticity—the assumption that firms respond to exogenous shocks and changes in competitive conditions. The centerpiece of economic microtheory is the deduction of autonomous responsiveness (mediated by self-interest) to changes in prices, technology, taxes, etc. Yet the truth is that firms change only with difficulty. Changing strategy and the structural forms and administrative procedures that undergird strategy is difficult, costly, risky, and time consuming.

The lack of plasticity is frequently called inertia—the strong persistence of existing form and function. A widely perceived example of organizational inertia is General Motors. Once the world leader in automobile production efficiency, General Motors was eclipsed in this regard by leading Japanese manufacturers during the 1970s. Senior management understood the nature of the productivity gap by 1979 yet despite a joint venture with Toyota in which world-class methods were used, the company has been unable to substantially change its overall productivity. In fact, many GM plants became less productive during the 1980s while Chrysler and Ford made broad and significant gains. Clearly, the basic problem facing the senior management of General Motors is not product-market strategy, but organizational change. Their challenge is not really competition, but their own inertia.

The fact of organizational inertia is not simply an "implementation" problem. If firms lack plasticity then the formulation of product-market and corporate strategy is itself fundamentally altered:

- Good product-market strategy must take into account a firm's inertias and not create new inertia without sufficient reason.
- The important strategy problem facing a firm may well be internal inertia rather than product-market conditions.
- Leaving inertia out of an analysis underestimates, perhaps drastically, the payoff to strategic change and innovation. Modern economic analysis of strategy presumes alert responsive competitors who will compete away a firm's profits unless there are the protections afforded by property rights, reputation, tacit know-how, or other strategic resources and isolating mechanisms. But if competitors are subject to inertia, this analysis is incorrect. Some of the great strategic success stories are due as much to the (temporary) inertia of competitors as to the cleverness of the

innovator (e.g., Timex, Federal Express).

A basic question motivating much of the research into the resource-based point of view was "Why are firms different?" The question which I suggest requires equal attention is "What are the sources of organizational inertia?" Or, "Why is change so difficult?"

Inertia and Evolution

The inertia of firms is a dominant theme in two fields loosely coupled with strategy: organizational ecology and evolutionary economics. Main-line organizational ecology assumes that firms are immutable (once born) and that populations of firms change over time through processes of birth, selection, and death. In evolutionary economics there is the additional theme that firms change, though not always adaptively, as occasional innovation and trial-and-error processes affect their basic work-routines and policies.

Within organization ecology Hannan and Freeman [1984] define inertia as change that is slow relative to the environment, so that a firm that is considered flexible in one setting might still have too much inertia to compete in a still faster-changing industry. Hannan and Freeman argue that much inertia is the unavoidable consequence of specialized investment in physical assets and social structures. In particular, they argue that these specialized investments meet the organizations need for *reliability* and *accountability*—the need to repeatedly produce the same type of product or process and the need to justify and explain process and choice to outsiders. They identify reproducibility with institutionalization, which gives "an organization a taken-for-granted character such that members do not continually question organizational purposes, authority, relations, etc." [1984:154]. Hannan and Freeman follow other sociologists in arguing that organizations have "cores" which are very difficult to change relative to more peripheral elements. The four core elements they identify are (1) stated goals, (2) the firm of authority, (3) the basic technology, and (4) the marketing strategy.

Within evolutionary economics the explanation for inertia is rooted in bounded rationality, routines, and tacitness. Nelson and Winter [19??] argue that the skills and capabilities of organizations are bound up in their routines for accomplishing tasks. These routines, *and the fact that they are routine*, constitute the skill set and the memory of the firm. Within this framework, inertia is the natural state of affairs—firms can only do what they have routines for doing. Lacking a routine for a new task, the new task does not really get done.

These perspectives are both helpful in understanding inertia. However, neither attempts to be comprehensive or analytic. In the next sections I begin to build a framework that attempts to encompass broader managerial view of inertia and begins to allow analysis of the constellation of inertial forces at work in different situations.

The Contexts of Change and Inertia

The problem of inertia arises when there is a need for change. This need may be internally induced (innovation) or externally compelled (competition). The change contexts facing firms can be usefully divided into those in which operating efficiency is primary and those in which the adroit utilization or reallocation of resources is primary.

Figure 1 is a guide to various change contexts. The horizontal axis (capability) measures the relative efficiency of the firm at performing its tasks. The vertical axis (fit) measures the adequacy of these tasks in meeting product-market demands. The ideal firm is both capable and fit (adapted). The recent problems at General Motors, I would argue, stem from a decline in relative capability. The basic strategy of the firm, its overall product line, its distribution system, its essential product-technology, is much less of a problem. Thus, GM is a case of decline in capability rather than fit and the problems of inertia it faces are those associated with developing and learning new or improved capabilities.

By contrast, IBM remains a very capable manufacturer of main-frame computers—its capabilities are excellent. Its recent problems stem from the increasing lack of fit between these capabilities and the environment. Market demands are less for main-frames and more for very inexpensive PCs networked into work-groups. IBM's fit with this new world is poor. Thus, the inertial problems IBM faces are quite different from those facing GM. Its management must deal with moving resources away from activities that are clearly being done well (on a technical basis) and which have been, in the past, the wellsprings of the firm's success. Another firm clearly in this position is Northrop, the maker of the Stealth Bomber. Its capabilities at its historic mission are superb, but the mission may well be obsolete.

The Five Frictions

With a nod to Michael Porter, I have organized the main sources of inertia—the main impediments to change—into five groups, called the *five frictions*:

1. Perceptual Distortions
2. Motivational Dampers
3. Analysis and Choice Paralyzers
4. Vested Values and Interests Blockades
5. Culture and Capability Disconnects

In many cases the components of these frictions are well known or easily comprehended. In other cases the issues are novel or subtle. In what follows I shall pass swiftly over many straightforward issues in order to concentrate on those that are novel or complex.

Perceptual Distortions

Change involves a set of activities beginning with perception. If perception is distorted, then change may be impeded. The fundamental sources of perceptual distortion inducing organizational inertia are:

- Myopia
- Lack of a Synthetic View
- Denial
- Grooved Thinking

Myopia. A firm suffering from myopia is unable to look into the future with clarity. Various individuals within the firm may be aware of the future consequences of current action or inaction, but the organization, taken as a whole, acts as if only the short-term matters. Myopia is most likely in a firm that has employed high-intensity short-term controls or in a firm which is "forgetful." In the former case, individuals bias their allocations of attention and effort towards problems that effect current

performance.¹

When individuals do not expect the organization to "remember" the connections between current actions and future results. In such cases, exacerbated by mergers, reorganizations, frequent personnel changes, etc., individuals have no realistic expectation that future results will be attributed to their current actions or decisions. The somewhat subtle consequence of this expectation is that senior management will behave myopically, expressing a rational disbelief in the claims of future threat or opportunity expressed by lower level managers.

A third source may be the failure of planning systems to counteract normal human myopia. Empirical evidence suggests that people's time-discount functions are steeper-than-exponential, leading to time inconsistent behavior.² Well-functioning organizational systems, it may be argued, help counteract short-term impulsiveness. And, by contrast, poorly functioning planning and review systems fail in this regard, producing organizational myopia.

Lack of a Synthetic View. Organizational perception is distorted when there is ineffective integration among the information and judgments available within the firm. Extreme decentralization can lead to this situation as can strong barriers to the flow of information between units. The lack of a synthetic view is most frequently observed when strategic planning is budget-driven and focused on forecasting.

Denial. A serious source of perceptual distortion is denial—the rejection of information that is contrary to what is desired or what is believed to be true. For years General Motors refused to admit that Japanese firms had higher absolute levels of productivity, denying the existence of the problem. A related syndrome is information filtering—the selective rejection of information that is unpopular, unpleasant, or contrary to doctrine. During the Vietnam war, US commanders defined the conflict as a war of attrition which was being won over time and systematically withheld from Washington information which contradicted their vision.³

Grooved Thinking. Janus [1972] has described as "groupthink" the restricted thinking that groups impose, punishing or rejecting ideas and information that deviate too much from orthodoxy. A somewhat different perspective is provided by Margolis [1987] who views patterns of thinking as mental habits whose structure and function are the same as physical habits. Finally, a third type of grooved thinking comes from the use of the "wrong" metaphor. Just as policy makers may struggle to decide whether the situation in Bosnia is "another Kuwait" or "another Vietnam," a metaphor, once accepted, provides a powerful restriction on future thought.

Motivational Dampers

Even if perception is accurate, organizations may resist change because the need is not felt with sufficient sharpness. The lack of sufficient motivation may be rational, or it may reflect agency or psychological problems. The most important motivational dampers are

- Direct costs and risks of change

¹Holmstrom and Milgrom [1990] provide a clever agency model of the allocation of effort among jobs, concluding that most effort will be allocated to that which is measured most accurately.

²See Postrel and Rumelt [1992] for a full discussion.

³See Sheehan, Neil. *A Bright and Shining Lie*. New York: Vantage, 1988.

- Cannibalization costs
- Cross subsidy comforts
- Leadership commitment to the status quo

Direct costs and risks of change. It is likely that change increases the risk of organizational failure (mortality), disrupts operations, and involves a great deal of expensive effort. Even more importantly, change may imply the abandonment of costly sunk specific investments. If these considerations apply to the firm as a whole, they are rational impediments to change. Applied to individuals or groups, they point to agency problems.

Note that an impediment to change may be rational. When Timex created the disposable watch, there were no imitators for ten years. Some of the inertia was surely wasteful, but some may have been rational. Were a quality Swiss manufacturer to produce disposable watches it would risk damage to its reputation. Similarly, whereas firms making new investments may be able to justify an expensive new technology, a firm with adequate though less efficient capacity may not be able to economically justify switching to the newer technology.

Cannibalization costs. Cannibalization problems occur when there is an asymmetric response to a company's action. For example, when interest rates rose sharply in the late 1970s, many banks and S&Ls began to offer new money-market accounts and other innovative services. However, those institutions with the largest volume of savings account deposits (paying 5 percent interest when market rates were 17 percent!) were least likely to market the new products. They did, of course, lose some customers to other banks. But they also reasoned that they would lose even more savings account customers were they to market the new products heavily themselves—the new products would cannibalize the very high profits being earned on the old products.

Cross Subsidy Comforts. The motivation to change is inhibited when a problem business is bundled with a profitable one. The rents from the profitable operation subsidize losses in the problem area, diminishing the sharpness of the incentive to change.

Leadership Commitment to the Status-Quo. It is a commonplace in business that radical change in strategy or structure doesn't happen without a change in leadership. Leaders may inaugurate change, but that change itself then becomes the new status quo.

Leadership is committed to the status-quo for several reasons. The first is simple ego-involvement. If the current leadership were the architects of the status quo, then they will feel a special attachment to it as well as having a deep appreciation for its (original) functionality. The second reason is that the status-quo is known whereas new structures and strategies are not. As Burke pointed out in his critique of the French revolution, new and yet untried systems generally have terrible and as yet unimagined flaws. The third and most subtle reason is that leadership, by its nature, involves commitment. The leadership of an organization must be the guarantor of explicit contracts, both internal and external. In particular, organizational members give energy and their own commitment to broad goals only to the extent that they perceive that the leadership itself has made a strong commitment to a particular set of priorities and ways of doing things. Major change requires a leader to repudiate prior commitments, thus lessening his or her ability to lead in the future. Hence,

entrenched leadership is normally a source of inertia.

Analysis and Choice Paralyzers

If perception is acute and motivation sharp, change may still be blocked by other forces. In particular, it may be difficult for the organization to choose a direction out of their difficulties. The impediment may be in the analysis of the situation or in choice itself. The major categories of friction in this area are:

- High speed and complexity
- Blocked belief system
- Departmental politics
- Ineffective vision

High Speed and Unfamiliar Terrain. In the 1970s John Boyd, a captain in the U.S. Air Force, concluded a study of why U.S. aviators had been so successful in air combat in Korea.¹ He found that although the MiG-15 was faster, quicker in the turn, and could climb faster than the U.S. F-86, the critical factor was that the F-86 provided better visibility and faster control. Thus the U.S. pilot could remain a step ahead, maintaining the initiative. Most importantly, Boyd found that once the MiG pilot found himself outmaneuvered, he froze or panicked.

Analysis is blocked or frozen when things happen too fast. If a competitor can go around the "Boyd loop" of observation, orientation, decision, and action faster than an opponent, the opponent may not simply struggle along, he may freeze-up or collapse. The "Boyd loop" phenomena is one form of what is known to everyone as "having the initiative" in a game.

When the decision situation is very complex, there may also be a similar blockage. Thus, for example, U.S. firms all dropped out of the liquid crystal flat-screen technology race when Japanese firms jumped too far ahead. The pace of events overtook their resource allocation systems and the complexity of the judgments to be made exacerbated the situation.

Blocked Belief System. The case of a leadership group wedded to the status quo has already been discussed. Analysis is similarly blocked when the leadership group's commitment is to a pattern of causal logic rather than to the status quo per se. Thus, a management group that believes "this business is about merchandising," can fail to respond, perhaps for decades, to a competitive attack in the form of improved logistics.

A more interesting problem of blocked beliefs arises when there is disagreement among the leadership group. Figure 2 illustrates the classic Condorcet paradox. Here three managers face three alternatives (downsize, merge, and R&D), and each ranks the three differently. The problem here is that the group's preferences are intransitive. If any of the three alternatives is considered the status-quo, there is always an alternative preferred to it by a 2-to-1 majority. Here "R&D" beats "downsize," "merge" beats "R&D," and "downsize" beats "merge." This type of paradox led Arrow [1963] to his "impossibility theorem" and has inspired a stream of related work. It all points to the fact that traditional concepts of "public interest" or the "organization's goals" are probably empty of meaning unless there is uniformity of opinion.

¹See Fallows [1981] and Smith [1985]. Thompson [1992] provides a business oriented interpretation.

What happens when these managers are asked to reach a "consensus?" Modern political theory establishes that here (and in general) there is no "rational" outcome, and that any of the outcomes can be obtained by manipulation of the agenda. My belief is that there are four basic patterns that can emerge. First, a leader may simply impose a decision, eliminating the illusion of group choice. Secondly, the choice process may cycle for some time without generating an outcome. Thirdly, the managers may recognize the "irrationality" of the situation and withdraw from useful participation (essentially colluding to avoid choice). Finally, other considerations or "games" may change the relative influence the managers have, resolving the paradox.

When managers disagree, there is no "rational" way to combine their beliefs. If speed is importance, leadership may have to abandon group decision processes. When process is important, and beliefs differ, inertia may well be the outcome.¹

Departmental Politics. This is one of the most obvious sources of inertia and little more need be said. Managers rarely act to unseat themselves or to terminate their own departments. Yet change inevitably involves winners and losers; some people and departments will gain resources and prestige and others will be lessened. Change will be fought by those who will clearly lose thereby and departmental wrangling over who will win and lose can slow change to a crawl.

Ineffective Vision. Even when analysis and choice have not been blocked, the direction chosen and especially its articulation may be so flawed that change is blocked. Vision (direction) may be ineffective because it is hypocritical. Hypocritical vision is dishonest, claiming values and goals that are known to be false. Announcing "We are a community . . ." is a lie for a firm that is about to lay off one-third of its workforce. Managers know when goals of "quality" or of being "ecologically sound" are hypocritical and cannot help but treat the rest of the vision with cynicism.

Vision can be ineffective because it is unresponsive to the clear challenges facing the firm. An unresponsive vision can hardly inspire commitment and change. Finally, vision can be ineffective because there is no trust in senior management's commitment to the vision. If people believe that this "vision" is simply "today's" plan, only to be replaced by another tomorrow, there will be no willing followers. If managers are to commit their energy, careers, time, and attention to a program of change, there must be trust that the direction chosen will not be lightly altered.

Vested Values and Interests

The fourth friction is the existence of vested values and interests. I distinguish this from simple politics in that here individuals and departments are taken to have strong emotional or value attachments to products, policies, or ways of doing things. These vested values and interests can easily be the greatest impediments to change a firm faces.

For many years a central concept in the strategy field was the experience curve—the idea that the cost of doing something falls as cumulative experience (production) increases. By analogy, I suggest that there is a vested value "curve:" the

¹Many group processes for reaching "consensus" have little to do with the content of the decision, but instead focus on getting each manager to "buy into" and "take ownership" of some overall plan. The objective is not to combine knowledge or balance beliefs, but to eliminate diversity and disagreement.

more one does a task the more one sees value in that activity. I am careful not to say one "likes" the activity, for that is another issue.

The psychology of vested value is dissonance reduction—the repetition of something difficult makes sense if it is valued. If one has been rewarded for doing the task, it will not only be valued, but liked as well.¹

Vested values lie at the heart of institutions. The defenders of vested values are usually the informal leadership network—the defenders of the society and of its norms. The paradox of change is that these same people, perhaps the best and the brightest, easily become the source of inertia. The problem of vested values is not with simple foot dragging, but with the organizational equivalent of patriotism.

Action Blockades and Disconnects

The fifth source of friction concerns those forces which prevent action. Even if perception has been sound, analysis and choice have proceeded, and the problem of vested values overcome, there may still be no change. The basic reasons for action blockades and disconnects are

- Leadership inaction
- Process inertia
- Collective action problems
- Capabilities gaps

Leadership inaction. For change to begin, the leadership must articulate a vision for change, must alter incentives, must take direct action where possible, and must shift power. If it fails to do these things, change will be inhibited. The need for direct action is perhaps worth emphasizing. The concept of leadership comes from the military notion of being "in front," of providing courage and energy by example. In business, the analog is for the senior management group to lead by example. If a new emphasis on service is the centerpiece of the strategy, and the marketing manager keeps on his old friend, the head of market research, despite a wide reputation for providing bad service to the product groups, there has been a failure of leadership by example.

Process Inertia. The life functions of a business are its processes—its ways of doing things. Complex processes possess great inertia. The knowledge of how certain steps are performed may be tacit, no one may have a complete understanding of the process, and the various routines that make up the process take on the force of habit. From a purely economics perspective, organizational change only requires a change in incentives. However, the habitual patterns of work have an inertial force that can be much stronger than any practical incentives.

Process inertia arises not only through the attraction of the habitual, but also because change may mean doing things for which the organization has no process experience. If change simply requires substituting one process for another, things are relatively straightforward. But change which takes the organization beyond its repertoire of routines is difficult and may well stall. In general, one cannot ask an organization to do something which has not been reduced to a routine. If novel routines are required, then macro-routines for creating novelty, with all their slowness and cost, are required.

Collective Action Problems. Action can be blocked by a variety of collective

¹Many Marines value boot camp without liking it.

action problems. The simplest is the first mover problem: if senior management has called for new initiatives, does it pay to be a first-mover? In many situations the incentives are clearly in favor of waiting to see how the first mover does. In such situations, the equilibrium is for no one to move at all. There are also analogous problems of free-riders that inhibit change even when first-movers have led the way.

The more complex collective action problem is that best described as "cultural." A dysfunctional culture may block change and itself be virtually impervious to alteration. Consider, for example, the simple question of cooperation (coordination) versus competition among departments. As a step towards a model, suppose that individuals in an organization meet in pairs and must each decide whether to cooperate ("C") or compete ("X"). Each individual has information that the other does not and each controls some resources that the other does not. Cooperation means try to act for the good of the company as a whole. In particular, it means making claims on the other person's resources that are justified by one's private information and accepting the other person's claims on one's own resources as valid. Competition means trying to act (covertly) for the good of the local unit. Concessions from the other are sought that will benefit one's own unit.

To make the game a bit concrete, example payoffs are shown in Figure 3. The net payoff to each player are symmetric and are simply a weighted average of the local and corporate payoffs. Figure 4 shows that if enough weight is placed on corporate payoffs, the equilibrium in the simple single-meeting game is CC. If local payoffs dominate, the equilibrium is XX. At intermediate levels of weighting, there are two pure-strategy equilibria: CC and XX. That is, there are two possible cultures and theory has little more to say about which will exist.

Suppose a firm had, over time, developed a culture having equilibrium XX. How might management move it to CC?¹ If a charismatic leader were able to convince the group that it was truly at CC, then the equilibrium would indeed change.² Another strategy is to break off a small unit of the organization and focus resources and attention on changing its culture. Then, the old organization must be slowly recombined with the new culture in such a way as to preserve the new rather than overwhelm it with the old. A third strategy would be to merge with a firm having culture CC and hope that the combination tilts in the right direction.

Because culture depends on mutual expectations, it is not easily changed. A culture that resists change or that does not fit the direction the firm needs to take can be an insurmountable source of inertia.

Capabilities Gaps. The final action blockade is simply a gap or disconnect between the tasks that need to be performed and the competencies and capabilities within the firm. A reasonable gap between reality and aspiration, called "stretch,"³ may motivate accomplishment. But too great a gap is discouraging and is more likely to inhibit than induce change.

¹Assume incentives are not sufficient and complete monitoring is not feasible.

²In this case one plays C if one believes that at least 60 percent of the others are playing C.

³See Hamel and Prahalad [1990].

Conclusions

If firms are not easily changed there are important implications for strategy. The overwhelming evidence is that organizations possess considerable inertia, yet strategy content models, including the resource-based view of the firm, tend to sidestep this issue. A complete "strategic" theory of the firm must deal with inertia.

Whereas the resource-based theory of the firm had the advantage of using a clear model of the firm as a foil, there is no clear simple model of management or organizational process. Hence, the study of inertia is complex and requires looking into a variety of disciplines. In this paper I have offered a number of ideas and conjectured garnered from economics, organizational sociology, cognitive psychology, political science, and general common sense.

Figure 1

DECLINE AND INNOVATION TRAJECTORIES

Figure 2
Condorcet Preferences

Rankings of Three Managers for Three Alternatives

	Manager		
	John	Jan	Jon
Downsize	1	2	3
Merge	2	3	1
R&D	3	1	2

Figure 3

Figure 4